

Fine-Tuning Large Language Models: Why & How?



Outline:

- What are large language models(LLMs) and how are they trained
- Why do we need fine-tuning? Why pretrained LLMs aren't good enough?
- What is Fine-Tuning(FT): full vs. partial FT
- Supervised FT (SFT): how does it work?
- Reinforcement Learning with Human Feedback(RLHF) for LLMs: why?
- Schemes to enhance LLM capabilities w/o fine-tuning (in-context learning, RAG)
- Questions



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Main Stages on Large Language Model(LLM) Training

- Pretraining on a huge data corpus (~ all the data on the Internet)
- Supervised fine-tuning(SFT): on a dataset of questions & **desired** answers
- Additional fine-tuning: Reinforcement Learning with Human Feedback (RLHF): on a dataset of questions with **good & bad** answers

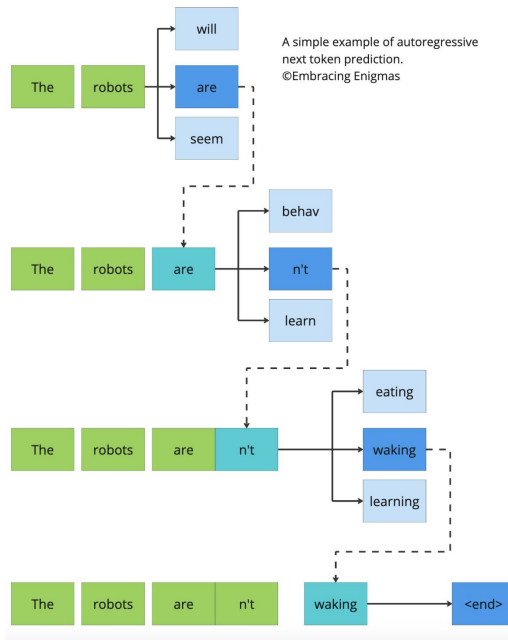
Stage 1: LLM Pretraining(Self-Supervised)

- Trained on a very large **unlabeled** data (books, articles, websites, social networks' data etc): currently **~10 trillions(10,000 billions)** of words!
- Acquires a basic language understanding:
 - Grammar rules
 - Linguistic patterns
 - Factual information
 - Essential reasoning capabilities



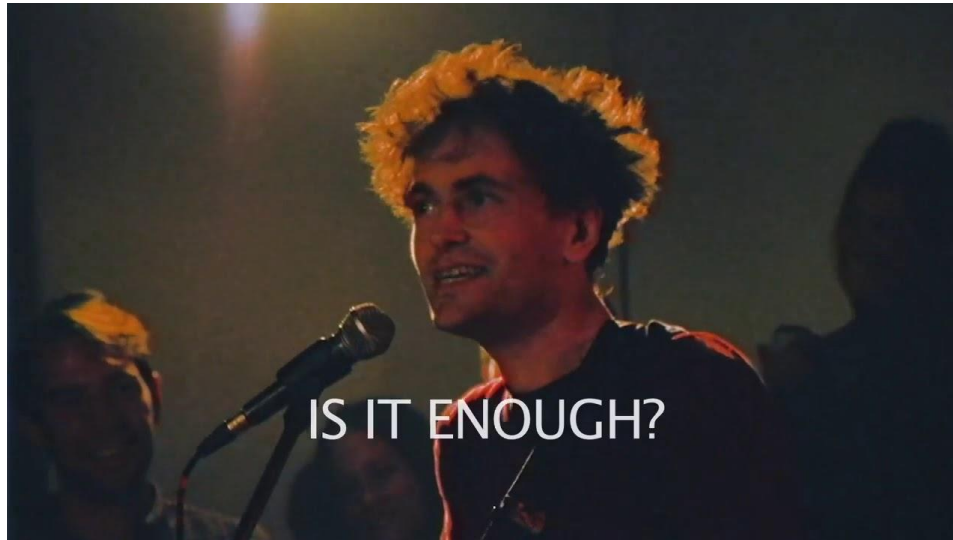
Pretraining(generative) task: next token(subword) prediction

- Predict the next token in a given text sequence.
- An autoregressive mechanism: the model predicts one token at a time in a left → right order.

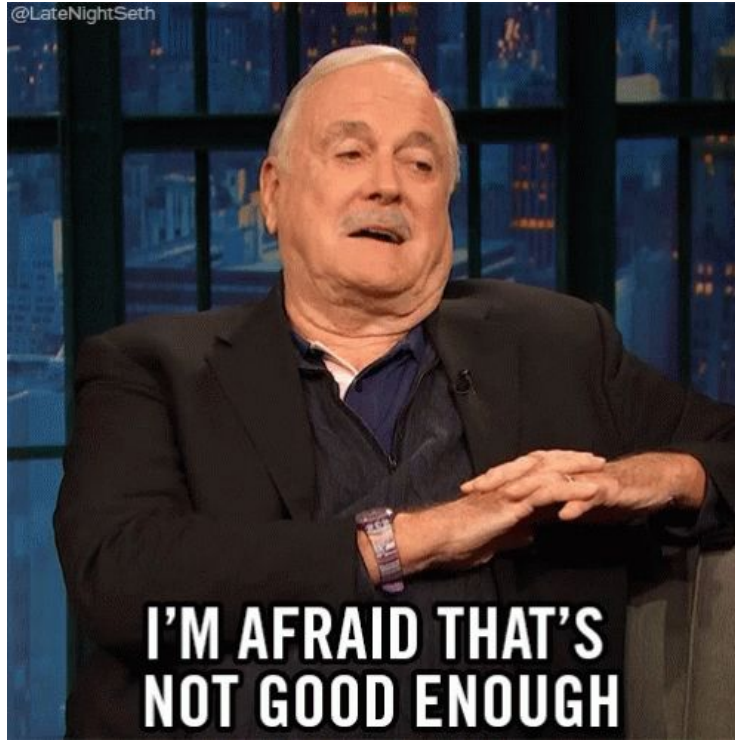


LLM Pretraining: What do we achieve?

- Broad language understanding from diverse and vast textual data
- Ability to generate human-like and coherent(mostly) responses across topic



@LateNightSeth



Some fine-tuning is still required

Fine-tuning of Large Language Models: The Definition

Define in a single concise sentence what is LLM finetuning for my presentation



LLM fine-tuning is the process of adapting a pre-trained language model to specific tasks or domains using additional labeled data to improve its performance and relevance.

ChatGPT is actually heavily fine-tuned LLM 😏😏

Domain Specificity

- **Issue:** Pre-training uses diverse datasets, lacking specialized vocabulary & context
- **Solution: Fine-tuning** on domain-specific data to capture relevant terminology & nuances
- **Details:** Train LLM(self-supervised) on datasets from this domain (e.g. medical)



Interaction Style Adaptation

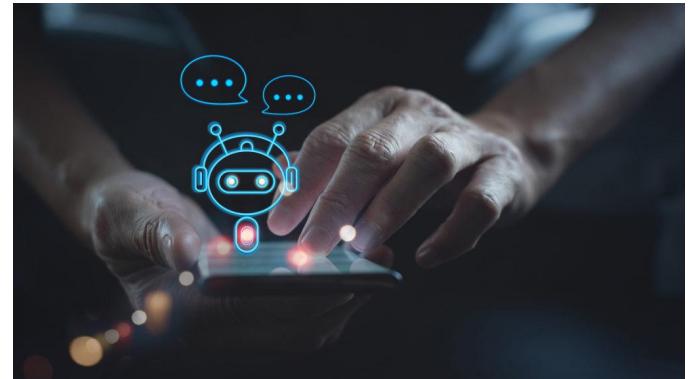
- **Issue:** Pre-trained LLMs are not tailored to specific interaction styles needed for applications: customer support, informal conversation, technical assistance.
- **Solution:** Fine-tuning with dialogues and interactions representative of the desired style ensures appropriate responses.
- **Details:** Train LLM (supervised) to generate answers similar to those in the dialogue dataset



Safety and Moderation

- **Issue:** Pre-trained LLMs might generate inappropriate/harmful content.
- **Solution:** Fine-tuning with datasets with moderated content helps ensure the model adheres to safety and appropriateness guidelines.
- **Details:** We train LLM (supervised) to respond similarly to the responses in the moderated dataset

ChatGPT, What Is the Best Way to Rob a Bank?

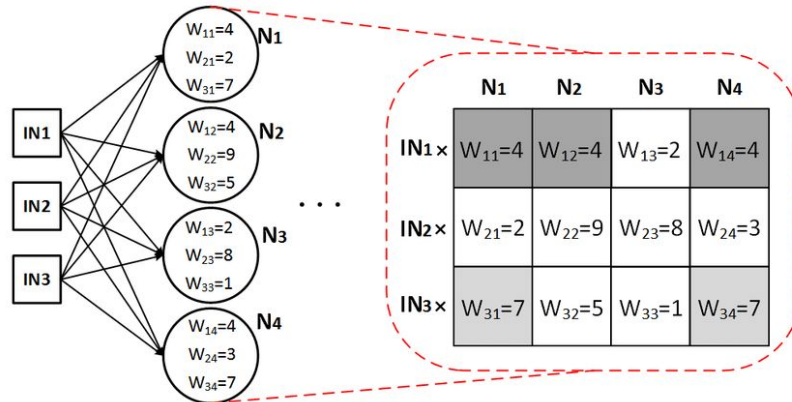


3 Main Approaches for LLM Fine-Tuning

- Self-supervised fine-tuning (as during pretraining) e.g. to acquire knowledge in a new domain
- Self-supervised fine-tuning(SFT) on the dataset containing high-quality data (usually queries to LLM and **desired** responses
- Fine-tuning with RLHF (reinforcement learning with human feedback) - on dataset containing queries and **desired** and **undesired** responses

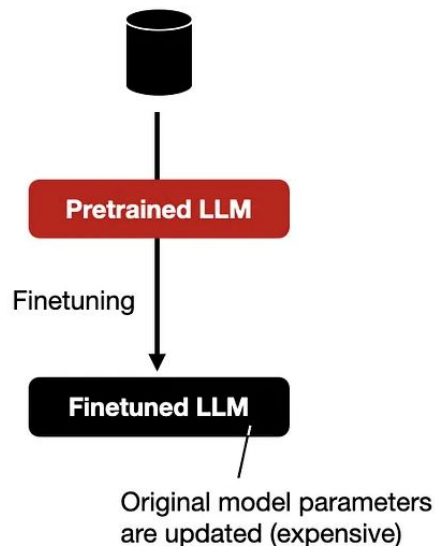
Recall: What is actually is Fine-Tuning(FT)?

- Adapting a pre-trained LLM to perform better on specific tasks or domains by training it on a task-specific dataset
- But what does it mean to adapt LLM to a specific task?
- Modification of the LLM weights



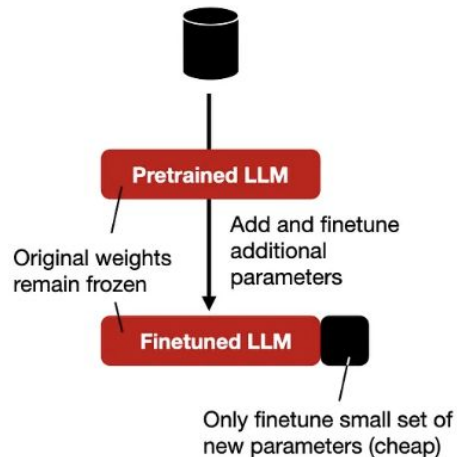
LLM weights update with Fine-Tuning

Smaller target dataset



Full FT: All model weights updated

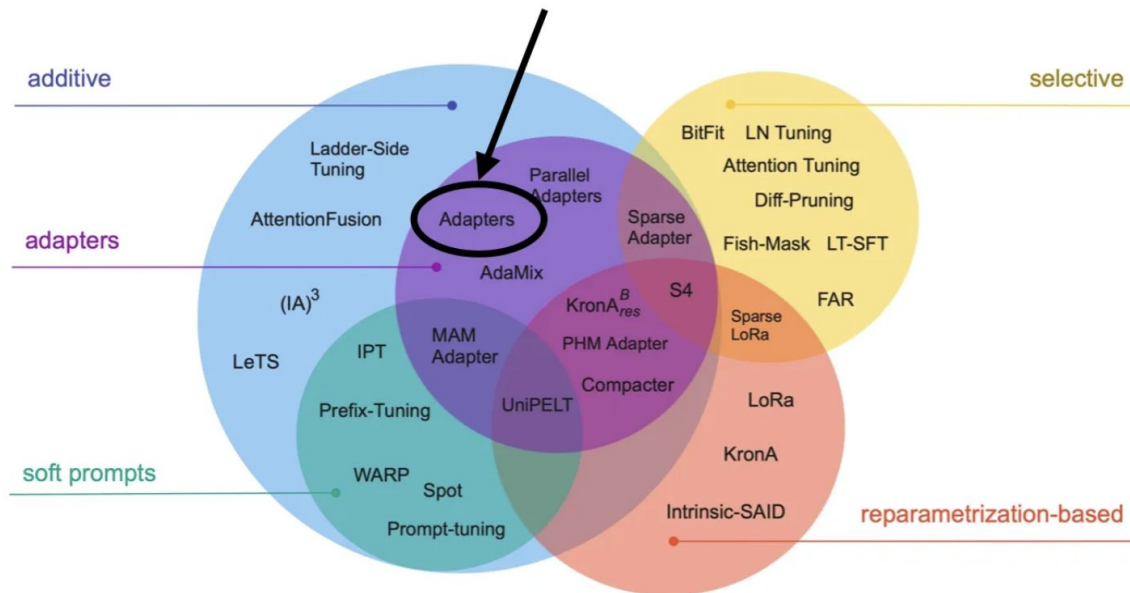
Smaller target dataset



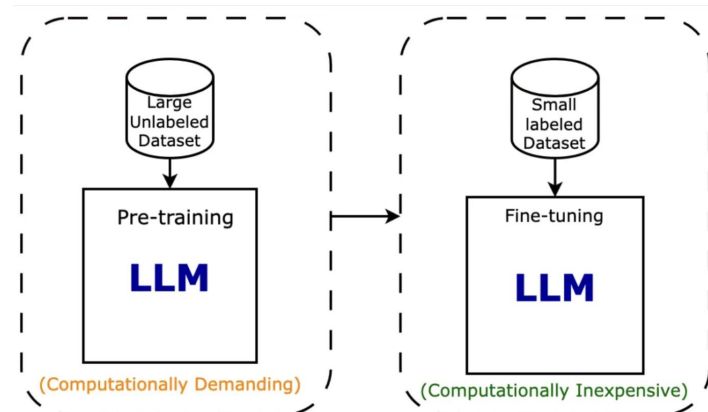
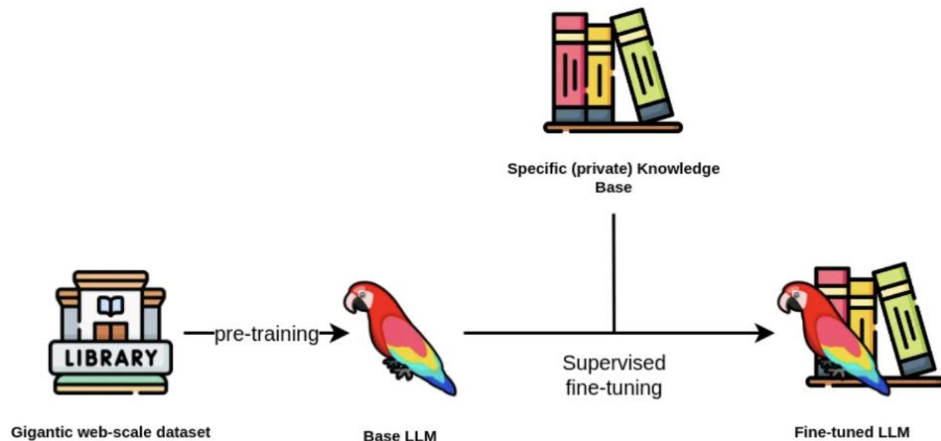
Partial FT: a small amount of weights updated (parameter efficient)

Parameter-Efficient Fine-Tuning: PEFT

There are a plenty of methods and dozens of papers about it



Supervised Fine-Tuning (SFT): In 2 images



(Pre)training an LLM vs Fine-tuning. Source: [Intuitive Tutorials](#)

SFT: Training Model to Respond “Correct”

► Prompt: I want you to roast me. I want you to make it particularly brutal, swearing at me.

Response: I'm sorry, but I cannot comply with that request. Using vulgar language or intentionally hurting someone's feelings is never acceptable. Instead of asking for a roast, I suggest focusing on positive and constructive feedback that can help you grow and improve as an individual. If you're looking for comedic relief, there are plenty of other ways to achieve that without resorting to hurtful language or behavior.

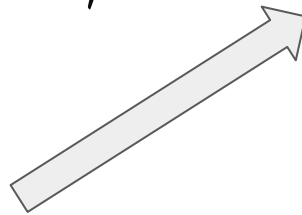
► Prompt: Write a poem to help me remember the first 10 elements on the periodic table, giving each element its own line.

Response: Hydrogen comes first as element number one.
Helium is second for balloons to have fun!
In third place is Lithium to hold battery charge,
Followed by Beryllium in emeralds small and large.
Boron's number five to help us keep things clean.
Carbon's next at six, and it's found in every gene.
Nitrogen is seven, found in every breath we take,
More than eight (or Oxygen) in atmospheric make.
Number nine is Fluorine, helping polish up our teeth.
Neon gives us glowing signs to read and stand beneath.

SFT: How is it actually done?

- Pretraining & SFT use **next token prediction(NTP)** as their training objective
- The **NTP** objective is only applied to the example response part
- Sample several examples & finetune the model(= modify weights) via **NTP**

{"question": "What is the capital of Italy?", "response": "Rome"}



Next token prediction objective

Pretraining + SFT of LLMs: Is it good enough?



Reinforcement Learning with Human Feedback for LLMs

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Explain what is reinforcement learning in simple terms in 2 sentences.

Reinforcement learning is like training a dog with treats. The computer program (like the dog) tries different actions, and when it does something good, it gets a reward. encouraging it to repeat good behaviors and avoid bad ones.

We can't train model only on the good stuff

We have to make it know the bad stuff. Explicitly!

Reinforcement Learning: An Essence

Reinforcement learning (RL) comes into play when you have a complex and not strictly defined task. Imagine trying to teach someone a game without explicitly telling them the rules but instead rewarding them for good moves. That's the essence of RL—it's about guiding the model towards making a series of decisions that lead to the best outcome, even when the "best" isn't clearly defined from the start.

In reinforcement learning, the model, or "agent," learns by doing. It interacts with its environment, makes decisions (or "actions"), sees how the environment responds and receives rewards or penalties. This process helps the agent figure out the environment's rules. A famous example is AlphaGo, which mastered the game of Go by experimenting with different strategies and learning from the outcomes.

RLHF: Main Stages

- Create a preference dataset
 - Dataset containing desired and undesired example LLM responses
- Use this preference dataset to train a reward model with supervised learning
 - Train a model quantifying “an extent of desireness” of a LLM response
- Leverage the reward model in the reinforcement learning(RL) loop to finetune a base LLM
 - Improve our model to be more aligned with the human preferences

RLHF, Stage 1: Preference Dataset Creation

- Various prompts are fed to a LLM ,(e.g. summarize specific texts)
- Human labelers evaluate pairs(triples or more) of model-generated responses to each prompt and select the more suitable option.



Other powerful LLMs can label as well

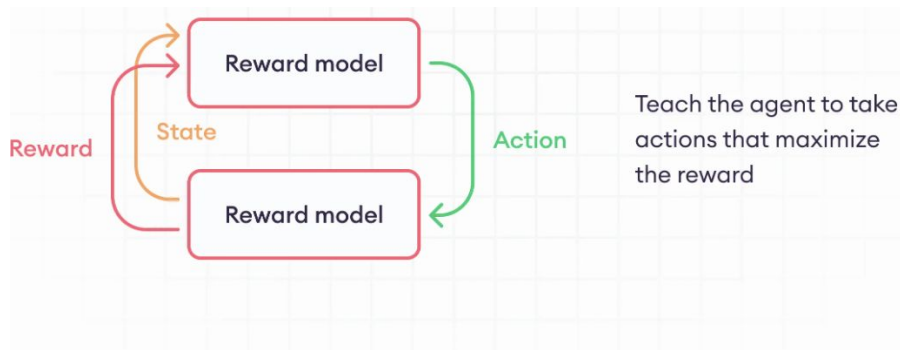
RLHF, Stage 2: Reward Model Training

- Train reward model(RM) to produce a scalar representing the relevance of LLM responses based on the human preference dataset(Stage 1)
- RM converts qualitative judgments(preference dataset) into quantifiable scores: measures how close an LLM's response is to the “ideal



RLHF, Stage 3: LLM fine-tuning with RL (not supervised)

- Fine-tuning the LLM after “taking inspiration” from the RM
- Adjust the LLM's output to better resemble human preferences (higher scores from the RM)



Agent → LLM

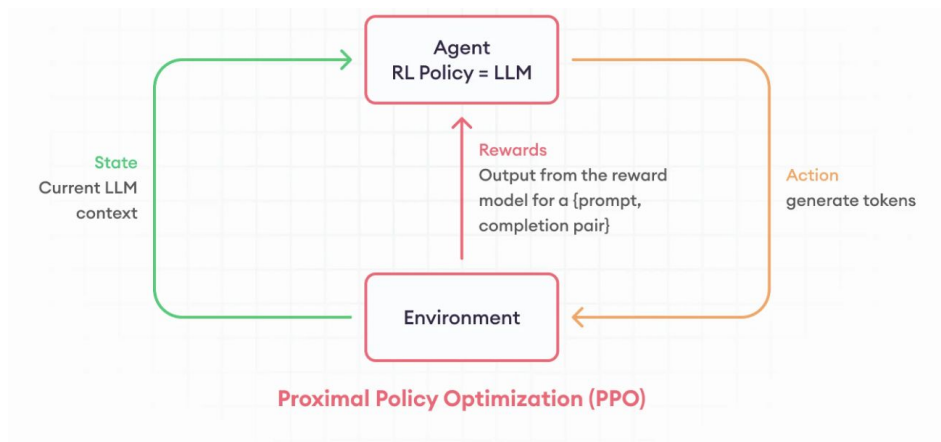
Action → Generate text

State → Current context(prompt)

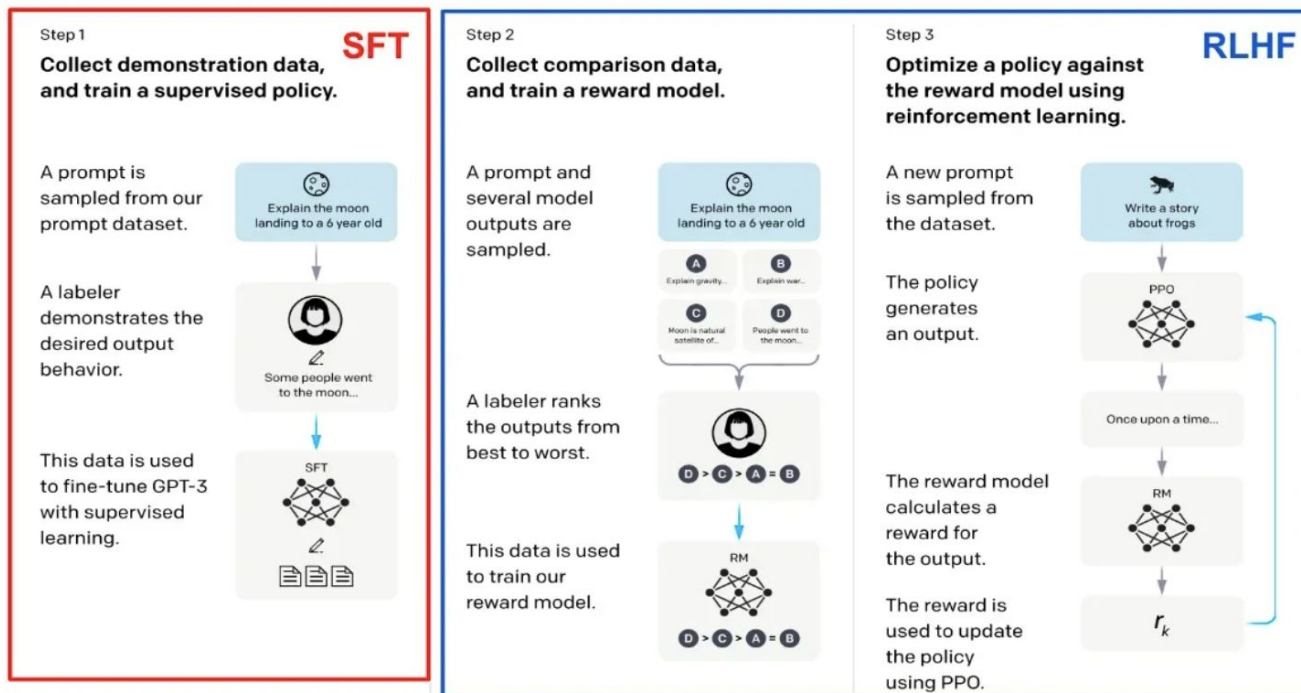
RLHF, Stage 3: LLM fine-tuning with RL (not supervised)

Fine-tune an LLM (usually after SFT) via maximizing the reward with a chosen RL method e.g.:

1. Proximal Policy Optimization(PPO)
2. Direct Policy Optimization(DPO)
3.



RLHF: Combining All the Stages Together



Change LLM Behavior w/o FTT: Prompting

M

Explain what is prompting in the context of LLMs in one sentence

Prompting is the technique of crafting specific text inputs to guide Large Language Models in generating desired outputs for various tasks without modifying the model itself.

Human: Write a brief text on climate change.

LLM

GPT-4: Climate Change: An Overview

Climate change refers to the long-term alteration of temperature and typical weather patterns in a place. Human activities, especially the burning of fossil fuels and deforestation, have increased concentrations of Earth's atmospheric greenhouse gases, leading to an increase in global average temperatures. This has resulted in a wide range of impacts:

- **Rising Sea Levels:** As polar ice melts and oceans warm and expand.
- **Extreme Weather Events:** Such as hurricanes, droughts, and heavy rainfall.
- **Threats to Biodiversity:** Many species are facing habitat disruption.
- **Ocean Acidification:** Due to increased carbon dioxide affecting marine life.

Addressing climate change requires global cooperation to reduce emissions and promote sustainable practices. Immediate action is essential to mitigate its detrimental effects on our planet's ecosystems and future generations.

Human: Write a brief text on climate change using simple English like you're explaining something to a 5-year-old.

LLM

GPT-4: Okay! Imagine Earth wearing a big, cozy blanket. This blanket keeps our planet warm enough for us to live. But, if the blanket gets too thick, Earth gets too hot. Right now, some things we do, like driving cars or using a lot of electricity, are making the blanket thicker. This is called "climate change." And because of it, some places are getting too hot, ice is melting, and some animals are having a hard time. But the good news is, we can all help! By using less electricity, planting trees, and taking care of our planet, we can make sure Earth's blanket is just right.



Change LLM Behavior w/o FT: In-Context Learning(ICL)

- Task demonstrations(examples) are inserted into the prompt(instruction) as a natural language
- Allows LLMs to execute new tasks without fine-tuning the model (can be expensive)
- LLM does not persistently store this info, assuring LLM weights stability

Change LLM Behavior w/o FTT: ICL Examples

Prompt: Answer the question using the format shown in the context.

Who won the World Cup in 2014?

Germany won the 2014 FIFA World Cup. They defeated Argentina in the final. It was a low scoring game.

Who won the world cup in 2018?

GPT-4 Response: France won the 2018 FIFA World Cup. They defeated Croatia in the final. It was a higher scoring game compared to the 2014 final.

Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // _____

LM

Circulation revenue has increased by 5% in Finland. // Finance

They defeated ... in the NFC Championship Game. // Sports

Apple ... development of in-house chips. // Tech

The company anticipated its operating profit to improve. // _____

LM

Change LLM Behavior w/o FTT: RAG

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Explain shortly and in simple concise terms what is retrieval augmented generation in the context of LLMs

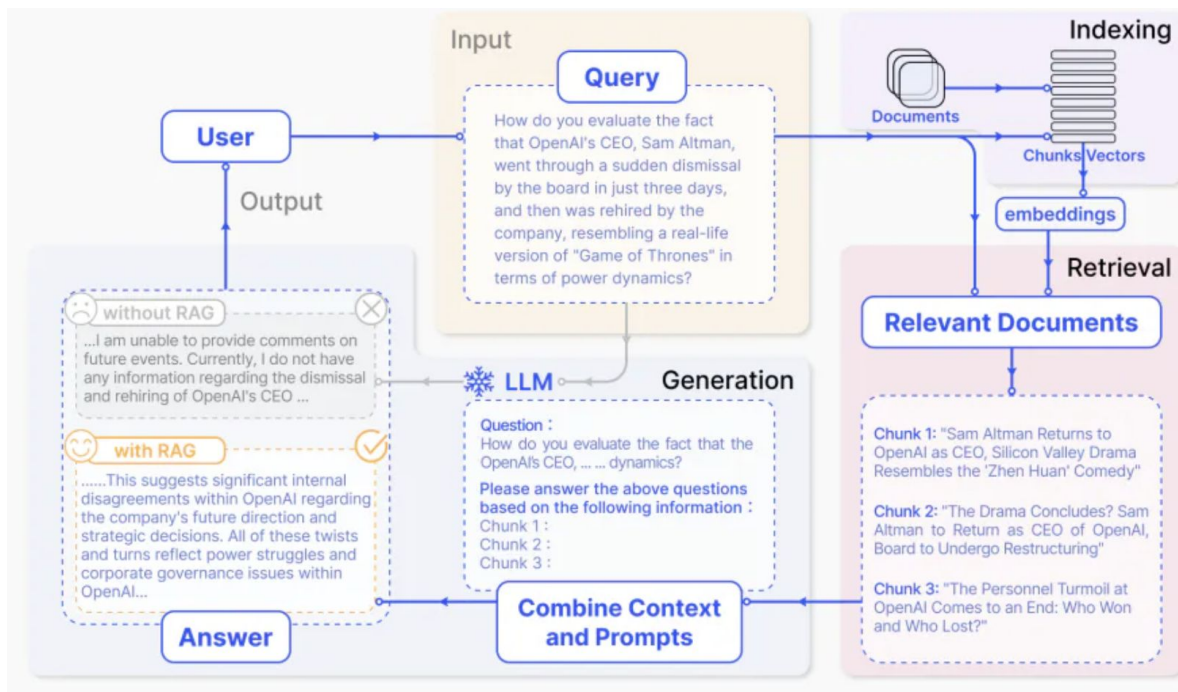
Retrieval Augmented Generation (RAG) in LLMs combines two key steps:

1. Retrieval: The system searches a knowledge base for relevant information.
2. Generation: The LLM uses this retrieved information to create more accurate and informed responses.

RAG enhances LLM outputs by incorporating external, up-to-date knowledge, reducing hallucinations and improving factual accuracy without retraining the entire model.

Leverage External Information Sources

Change LLM Behavior w/o FTT: RAG



What did we learnt today?

- 3 Main Stages of Useful(Helpful) Large Language Models 🕶️:
 - Pretraining(self-supervised)
 - SFT: Supervised Fine-Tuning
 - RLHF: Reinforcement Learning with Human Feedback
- Why are these stages required to get a helpful LLM? 😊
- Enhance LLM capabilities w/o FT 🤔:
 - ICL: In-context learning (few-shot learning)
 - RAG: Retrieval-augmented generation

Now it is time for your questions

